

977. Squares of a Sorted Array

Easy Topics Companies

Given an integer array `nums` sorted in **non-decreasing** order, return an array of **the squares of each number** sorted in **non-decreasing** order.

Example 1:

Input: `nums = [-4,-1,0,3,10]`
Output: `[0,1,9,16,100]`
Explanation: After squaring, the array becomes `[16,1,0,9,100]`. After sorting, it becomes `[0,1,9,16,100]`.

Example 2:

Input: `nums = [-7,-3,2,3,11]`
Output: `[4,9,9,49,121]`

Constraints:

- $1 \leq \text{nums.length} \leq 10^4$
- $-10^4 \leq \text{nums}[i] \leq 10^4$
- `nums` is sorted in **non-decreasing** order.

Follow up: Squaring each element and sorting the new array is very trivial, could you find an $O(n)$ solution using a different approach?



Seen this question in a real interview before? 1/6

Yes No

Accepted 2,816,778 / 3.8M Acceptance Rate 74.1%

Topics

Code

Java Auto

```
1 class Solution {
2     public int[] sortedSquares(int[] nums) {
3         int n = nums.length;
4         int[] result = new int[n];
5
6         int left = 0;
7         int right = n - 1;
8
9         // Fill result from largest to smallest
10        for (int i = n - 1; i >= 0; i--) {
11            if (Math.abs(nums[left]) > Math.abs(nums[right])) {
12                result[i] = nums[left] * nums[left];
13                left++;
14            } else {
15                result[i] = nums[right] * nums[right];
16                right--;
17            }
18        }
19
20        return result;
21    }
22 }
23
24
```

Saved

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Testcase Test Result

Input

`nums =`
`[-4,-1,0,3,10]`

Output

`[0,1,9,16,100]`

Expected

`[0,1,9,16,100]`

Contribute a testcase